

Driverless e-vehicles in IIT from 2024

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Chennai: Imagine getting into a driverless electric vehicle, keying in the destination and getting dropped off in a matter of minutes – this will become a reality inside the Indian Institute of Technology, Madras, in 2024.

Electric vehicle 'bolt' will replace buses inside the campus by 2025. 'Bolt' has been developed by Team Abhiyaan comprising a group of multi-disciplinary students of IITM, committed to building intelligent and safe ground navigation systems.

Mugdha Meda, a second-year mechanical engineering student and member of the team, said the heart of the vehicle is the computer unit. "Otherwise, it is just a golf cart. The autonomous



INNOVATION: IIT-M Director V Kamakoti inaugurated the open house

vehicle can detect lanes, speed breakers, pedestrians and other vehicles with the help of cameras. It has electronic control for acceleration, brake and steering," she said.

Ananya Dhanvantri, a second-year engineering de-

sign student and member of the team, said 'bolt' also has an emergency stop button to automatically shut down the vehicle in case of a glitch. "Close to 94% of road accidents are caused due to human error. We are trying to

put an end to this," she said.

'Bolt' was among the 70 tech projects showcased at the Centre for Innovation (CFI) during the annual open house held on Sunday.

Another interesting project that was on display was the prototype of a solar car made by team Agnirath of IIT that works towards creating a sustainable future.

"The actual vehicle will have a glass fibre body and carbon fibre monocoque chassis. We will be taking this for the World Solar Challenge to be held in Australia this October. We will have to ride the fully solar powered car for 3,000 km from Darwin to Adelaide," said Bindusara Reddy, team leader.

The vehicle can travel at a maximum speed of 110 kmph.

"The vehicle is designed

in such a way that it is least air resistant compared to normal cars. It can cut through wind with ease," added Karthik Ruikar, another team member.

The CFI Open House 2023 featured innovative projects spanning diverse fields, ranging from 3D printing, astronomy and AI/ML to drones, autonomous vehicles, hyperloop and techno-entertainment.

Students had also displayed a soft robotics fish that is being developed to study aqua life and map the seabed.

"We will be fitting cameras and make it autonomous. It will be able to avoid obstacles, escape predators, etc. This will not cause much turbulence, hence it can be helpful for a detailed study," said Harsha Srinivas, one of the members of the project.